

Name: SOLUTIONS

GCSE (1 - 9)

Pressure



Buy me a coffee

Solutions by

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These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](https://www.pearson.com/uk/qualifications/edexcel/gcse/maths)

They contain exam questions from up until the **Summer 2023** exam series

More recent exam papers do exist but are withheld for mock exams

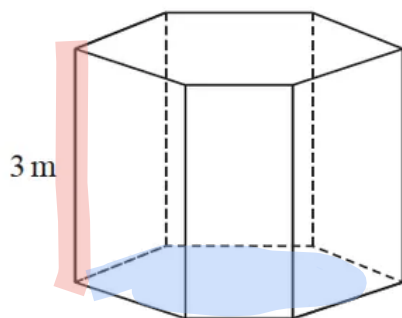
As new exam papers are released these worksheets will gradually be updated

The **most current** version of these worksheets will always be available at: [kennethstafford.co.uk](https://www.kennethstafford.co.uk)

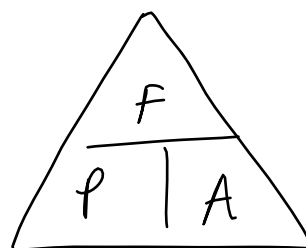
Question	Where the question appeared
1	Edexcel, GCSE, Autumn 2020, Paper 1F, Q27 Edexcel, GCSE, Autumn 2020, Paper 1H, Q8
2	Edexcel, GCSE, Summer 2018, Paper 2F, Q25 Edexcel, GCSE, Summer 2018, Paper 2H, Q6
3	Edexcel, GCSE, Summer 2023, Paper 1F, Q27
4	Edexcel, GCSE, November 2022, Paper 1F, Q25

1

The diagram shows a prism placed on a horizontal floor.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$



The prism has height 3 m

The volume of the prism is 18 m^3

The pressure on the floor due to the prism is 75 newtons/m^2

Work out the force exerted by the prism on the floor.

$$\text{Area of cross section} = \frac{18}{3} = 6 \text{ m}^2$$

$$\begin{aligned} \text{Force} &= \text{Pressure} \times \text{Area} \\ &= 75 \times 6 = 450 \end{aligned}$$

450 newtons

(Total for Question 8 is 3 marks)

[Edexcel, GCSE, Autumn 2020, Paper 1F, Q27
Edexcel, GCSE, Autumn 2020, Paper 1H, Q8]

2

A force of 70 newtons acts on an area of 20 cm^2

The force is increased by 10 newtons.

The area is increased by 10 cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Helen says,

"The pressure decreases by less than 20%"

Is Helen correct?

You must show how you get your answer.

$$\text{Pressure} = \frac{70}{20} = 3.5$$

$$\text{New Pressure} = \frac{70 + 10}{20 + 10} = \frac{80}{30} = 2.\dot{6}$$

$$\text{Decrease} = 3.5 - 2.\dot{6} = 0.8\dot{3}$$

$$\text{Percentage Change} = \frac{0.8\dot{3}}{3.5} \times 100 = 23.8\%$$

$$23.8\% > 20\%$$

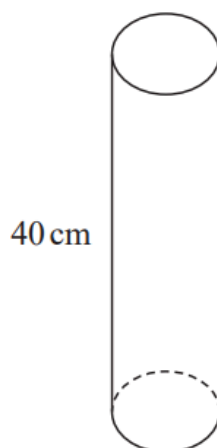
No. It decreases by more than 20%

(Total for Question 6 is 3 marks)

[Edexcel, GCSE, Summer 2018, Paper 2F, Q25
Edexcel, GCSE, Summer 2018, Paper 2H, Q6]

3

The diagram shows a solid cylinder on a horizontal floor.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{Area} = \frac{\text{Volume}}{\text{height}}$$

$$\text{Area} = \frac{1200}{40} = 30 \text{ cm}^2$$

The cylinder has a

volume of 1200 cm^3
height of 40 cm .

The cylinder exerts a force of 90 newtons on the floor.

Work out the pressure on the floor due to the cylinder.

$$\text{Pressure} = \frac{90}{30} = 3$$

..... 3 newtons/cm²

(Total for Question 27 is 3 marks)

[Edexcel, GCSE, Summer 2023, Paper 1F, Q27]

4



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

A storage tank exerts a force of 10 000 newtons on the ground.

The base of the tank in contact with the ground is a 4 m by 2 m rectangle.

Work out the pressure on the ground due to the tank.

$$\text{Base Area} = 4 \times 2 = 8 \text{ m}^2$$

$$\text{Pressure} = \frac{10000}{8} = 1250$$

1250

..... newtons/m²

(Total for Question 25 is 2 marks)

[Edexcel, GCSE, November 2022, Paper 1F, Q25]